

1. Smart Home Automation System: Intelligent Agent Analysis

A. Explanation and Functionality of Agent Types

- **Simple Reflex Agent:**
 - Mechanism: Selects actions based entirely on the current percept, ignoring the history of percepts, using condition-action rules.
 - Smart Home Functionality: Operates via direct triggers. For example, “If motion is detected, then turn on the lights.” It reacts instantly to immediate sensor inputs without retaining state.
- **Model-Based Reflex Agent:**
 - Mechanism: Maintains an internal state to track aspects of the world that are unobserved or changing over time, using a model of how the world works.
 - Smart Home Functionality: Remembers past states and trends. For example, it tracks whether the user prefers lights off when the room has been vacant for a specific duration, rather than just reacting instantly to a temporary drop in motion signals.
- **Goal-Based Agent:**
 - Mechanism: Operates using explicit goals describing situations that are desirable. It combines goal information with environmental modeling to choose actions that achieve the goal.
 - Smart Home Functionality: Plans actions to reach a specific target state. For example, “Adjust AC settings until the indoor temperature reaches exactly 22°C by 7:00 PM.” It evaluates multiple steps to satisfy that specific target condition.
- **Utility-Based Agent:**
 - Mechanism: Evaluates actions not just on whether they achieve a goal, but how well they satisfy a general preference or happiness metric (utility function) when multiple trade-offs exist.
 - Smart Home Functionality: Balances competing objectives like comfort, energy savings, and security. For example, it weighs the utility of lowering the temperature for comfort against the disutility of high electricity costs, finding the optimal balance rather than just a binary goal.

B. Justification of the Most Effective Approach

A **Utility-Based Agent** (or a **Hybrid Utility-Model-Based Approach**) is the most effective for this system.

- **Why:** A smart home cannot rely on rigid condition-action rules (Simple Reflex) or binary goals alone because comfort, energy efficiency, and security involve continuous trade-offs. For instance, cooling a house to maximum comfort maximizes comfort utility but minimizes energy efficiency utility. A utility-based agent calculates a quantitative score for various configurations, allowing it to adapt dynamically to user preferences, minimize power bills, and maintain optimal security simultaneously.

2. Unknown Maze Navigation: Search Strategies and Agent Design

A. Uninformed Search Strategies in Unknown Mazes

- **Uniform Cost Search (UCS):**
 - Mechanism: Expands the lowest-cost node first, defined by path cost $g(n)$.
 - Application: Finds the optimal path if step costs vary (e.g., rough terrain vs. smooth floor).
- **Iterative Deepening Search (IDS):**
 - Mechanism: Gradually increases a depth limit, repeatedly running Depth-First Search until the goal is found.
 - Application: Explores unknown trees or graphs level-by-level without requiring massive memory.

B. Advantages, Disadvantages, and Complexities

Search Strategy	Time Complexity	Space Complexity	Advantages	Disadvantages
Uniform Cost Search	$O(b^{1+\lceil C^*/\epsilon \rceil})$	$O(b^{1+\lceil C^*/\epsilon \rceil})$	Guaranteed optimal path for varying step costs; complete if step cost $\epsilon > 0$.	High memory and time consumption; expands all nodes in circles around the start if costs are uniform.
Iterative Deepening Search	$O(b^d)$	$O(bd)$	Memory efficient ($O(bd)$); optimal for unit step costs; explores exhaustively.	Redundant generation of states at each iteration; inefficient if branching factor b is very high.

C. Relation to Intelligent Agents and Decision-Making

- **Agent Design Impact:** In an unknown environment, a **Learning or Knowledge-Based Agent** is required. Unlike a simple reflex agent that would get trapped in dead ends, an agent tracking its explored states builds an internal graph representation of the maze.
- **Decision-Making:** The agent interleaves planning and execution. It updates its search tree dynamically as new corridors or walls are revealed by its sensors.

D. Most Effective Combination and Justification

- **Recommended Combination:** A **Learning Utility-Agent / Online Search Agent** paired with an **Online A*-style heuristic search** (or modified UCS with local learning, such as **RTAA***).
- **Justification:** Since the maze is unknown, offline uninformed searches like standard UCS or IDS cannot be executed upfront. An online search agent uses heuristic exploration to make informed local choices while mapping the maze, ensuring it finds an escape route efficiently without getting trapped indefinitely in blind alleys.